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**REPRODUCTION AND CONTROLLED
EVALUATION OF CRYPTOMAMBA
FOR NEXT-DAY BITCOIN PRICE
FORECASTING**

MASTER'S THESIS

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Abstract

This thesis reproduces CryptoMamba-v (CM-v) for next-day Bitcoin closing-price forecasting and reassesses the original paper’s findings. Under the 350-day protocol, the reproduced checkpoint’s RMSE, MAE, and MAPE differ from the reported values by 0.892%, 1.057%, and 0.718%, respectively. The controlled evaluation then compares CM-v, the experimental CMamba-T/S5-Full model, and naive persistence on the same 304 validation dates and 304 test dates. Their test RMSE values are 1,703.51, 1,660.83, and 1,657.55. Neither the Diebold–Mariano test nor the block-bootstrap interval confirms an RMSE advantage for S5-Full over CM-v. CM-v reaches 57.57% directional accuracy, but McNemar’s test does not confirm a directional difference between the two models. Persistence is excluded from this metric because it carries the current price forward. The paired RMSE and directional conclusions remain unchanged for bootstrap block lengths $L = 5, 7, 14$.

Two additional evaluation layers are provided. First, a zero-cost replay preserves the paper’s released trading algorithms for checkpoint reconciliation. Second, a separate self-financing engine charges transaction costs on traded notional, limits gross exposure, and reconciles final equity. Buy-and-hold exceeds every forecast-driven strategy at all three costs; at a 0.2% test cost, it ends at \$158.09. S5-Full has 57,249 parameters compared with 136,952 for CM-v. It achieves lower RMSE than CM-v on validation and test, but does not outperform persistence. Finally, a five-screen Console presents data, evaluation, checkpoint inference, trading, and architecture. Its primary workflow accepts a new dataset, selects either the CM-v or S5-Full checkpoint, and produces a forecast without retraining.

The application context is Vietnam’s pilot crypto-asset market under Resolution 05/2025/NQ-CP. The thesis does not provide legal analysis or investment advice.

Keywords: *Bitcoin, time-series forecasting, Mamba, research reproduction, naive persistence, forecast testing, self-financing backtest, Vietnamese crypto-asset market.*

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
BTC	Bitcoin
CM-v	CryptoMamba-v, the variant that includes volume
DM	Diebold–Mariano
HAC	Heteroskedasticity and Autocorrelation Consistent
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MDD	Maximum Drawdown
OHLCV	Open–High–Low–Close–Volume
RMSE	Root Mean Square Error
ROI	Return on Investment
SSM	State Space Model

Chapter 1

Introduction

1.1 Research Context

Next-day Bitcoin price forecasting is difficult because the price series is non-stationary, highly volatile, and strongly autocorrelated at the absolute-price scale. In this setting, a low error alone does not demonstrate that a model extracts new information: simply carrying the current closing price forward is already a strong control.

The CryptoMamba paper [1] applies Mamba [2] to daily BTC-USD data and reports that CryptoMamba-v outperforms LSTM-v, Bi-LSTM-v, GRU-v, iTransformer-v, and S-Mamba-v in RMSE, MAE, and MAPE. It also feeds forecasts into three trading algorithms. However, the published tables contain only aggregate metrics, without date-level errors for paired tests, and the trading evaluation does not model transaction costs through a self-financing ledger. This thesis addresses that evaluation gap while also implementing a lightweight variant and an inspectable explanatory system.

1.2 Objectives and Research Questions

The objective is to reproduce CM-v, evaluate forecasting and trading under defined protocols, experiment with a lightweight variant, and visualize the full workflow in a five-screen Console. The four research questions are:

- **RQ1 – Reproduction.** To what extent do CM-v and the released trading replay reproduce the paper’s results?
- **RQ2 – Evaluation.** On identical target dates, how do CM-v, S5-Full, and naive persistence differ in forecast error, direction, and statistical significance?
- **RQ3 – Trading.** How do forecast results translate into trading outcomes when the paper replay is separated from a corrected self-financing engine with costs?
- **RQ4 – Model.** Does S5-Full provide a lighter model with sufficient experimental evidence to support a forecasting advantage?

1.3 Scope

The study uses one asset, BTC-USD, with daily OHLCV candles, a one-step forecast horizon, and the time periods defined by the paper. The local controlled evaluation includes only three models with date-level predictions that can be aligned exactly: reproduced CM-v, S5-Full, and naive persistence. Other model rows appear only in the *Paper-reported* reference table and are not included in local paired tests.

As an application context, the project is situated within Vietnam’s five-year pilot crypto-asset market under Resolution 05/2025/NQ-CP, effective from 09-09-2025 [11]. The Resolution defines context only; the technical scope does not extend to legal analysis, compliance assessment, or investment advice.

The primary workflow is paper data or a new CSV → data validation → CM-v/S5-Full checkpoint selection → next-day price prediction. Historical replay is retained to illustrate past behavior. External

HTTP inference is optional rather than a prerequisite. The product is a research prototype, not an advisory or trade-execution system.

1.4 Contributions

1. **Reproduction.** Reproduce CM-v, reconcile the official and reproduced checkpoints, and audit the paper’s trading replay.
2. **Evaluation.** Add naive persistence, compare models on identical target dates, and test the observed differences using DM and Wilcoxon procedures. Uncertainty, chronological variation, and fee sensitivity are evaluated using block bootstrap, chronological halves, and three fixed fee rates.
3. **Model.** Implement and evaluate S5-Full as a lightweight CMamba-T experiment, reporting positive, negative, and non-significant results on their correct splits.
4. **System.** Build a five-screen Console for the data $\rightarrow$ prediction $\rightarrow$ evaluation $\rightarrow$ trading $\rightarrow$ architecture flow, with direct inference from two local checkpoints.

Chapter 2

Background and Related Work

2.1 Time-Series Forecasting and the Persistence Baseline

For one-step forecasting, naive persistence is defined as

$$\hat{y}_{t+1} = y_t, \quad (2.1)$$

where y_t is the current closing price. This baseline has no trainable parameters and issues neither upward nor downward signals. It tests whether a deep model improves on the most recent observation or merely reproduces the current price level.

iTransformer reverses the conventional Transformer tokenization scheme to model relationships across variables [3]. S-Mamba uses bidirectional Mamba for cross-variable relationships and a feed-forward network for temporal relationships [4]. Both architectures appear in CryptoMamba’s comparison table, but this thesis uses only their published aggregate metrics because date-level forecasts are unavailable.

2.2 State Space Models and Mamba

A linear SSM represents hidden state h_t and output z_t as

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad z_t = Ch_t. \quad (2.2)$$

Mamba makes state-transition parameters input-dependent, allowing information to be retained or discarded at each sequence element while maintaining linear complexity in sequence length [2]. The interpretation of a scan therefore depends directly on which tensor axis is treated as the sequence axis.

2.3 CryptoMamba-v: Published Description and Released Execution Path

The paper describes CryptoMamba as a sequence of C-Blocks followed by a Merge block. Each C-Block contains multiple CMBlocks and an MLP; the input is a window of past observations and the output is the next-day closing price [1]. Figure 2.1 separates this conceptual description from behavior that can be verified in the released source.

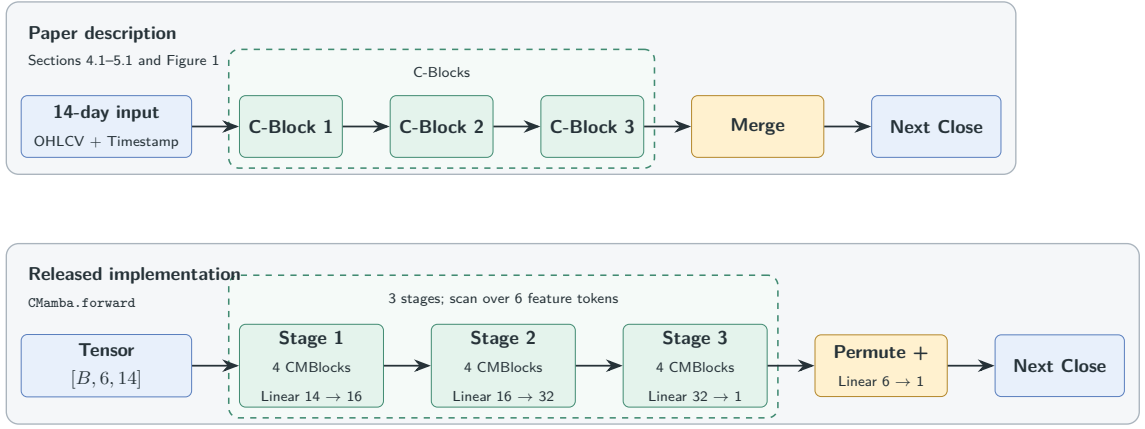


Figure 2.1 – Paper-described pipeline versus the source-derived CM-v execution path. The upper row combines the data/input description in Sections 4.1 and 5.1 with the architecture in Section 4.2 and Figure 1 of CryptoMamba; the lower row is derived from the released `CMamba.forward` implementation.

In the released code, CM-v accepts a $[B, 6, 14]$ tensor. The Mamba block receives six feature tokens, each with width 14. Three linear stages change the width as $14 \rightarrow 16 \rightarrow 32 \rightarrow 1$, and each stage contains four CMBlocks. The final head permutes the tensor and applies Linear $6 \rightarrow 1$. The executable configuration has 136,952 parameters. This observation describes only the released execution path; the paper’s conceptual diagram does not expose tensor-axis details.

2.4 Forecast Accuracy Comparison

The Diebold–Mariano (DM) test evaluates the null hypothesis that two forecasts have equal predictive accuracy [5]. This thesis uses the squared-loss differential $d_t = e_{A,t}^2 - e_{B,t}^2$ with a HAC variance estimate at lag 1 following Newey–West [7]. The Wilcoxon signed-rank test [6] is applied independently to the absolute-error differential $|e_{A,t}| - |e_{B,t}|$. These tests answer different questions and are therefore not presented as interchangeable measurements.

Chapter 3

Research Methodology

3.1 Data and Chronological Splits

Daily BTC-USD data follow the paper’s protocol from 17-09-2018 to 17-09-2024. Input fields are Timestamp, Open, High, Low, Close, and Volume. Splits preserve chronological order without shuffling (Table 3.1).

Table 3.1 – Data splits under the CryptoMamba protocol.

Split	Period	Role
Train	17-09-2018 to before 17-09-2022	Parameter estimation
Validation	17-09-2022 to before 17-09-2023	Checkpoint selection
Test	17-09-2023 to before 17-09-2024	Final evaluation

CM-v uses a 14-day window, whereas S5-Full uses a 60-day window. Paired comparisons use the intersection of target dates for which all three models produce forecasts: 304 validation dates from 16-11-2022 to 15-09-2023 and 304 test dates from 16-11-2023 to 14-09-2024. All local tables, tests, and corrected backtests use these 304 common target dates. The 350-day aggregate under the paper protocol is used only for RQ1 and is not mixed with the 304-day paired evaluation.

3.2 Models and Checkpoints

Both models are evaluated against the same final target: the raw next-day Close. Their input and output contracts differ as summarized in Table 3.2.

Table 3.2 – CM-v and S5-Full input, preprocessing, and output contracts.

Component	CM-v	S5-Full
Final target	Raw next-day Close	Raw next-day Close
Input window	14 days, $[B, 6, 14]$	60 days, $[B, 5, 60]$
Input fields	Timestamp + OHLCV	OHLCV; no Timestamp
Price input	Raw prices	$x/C_t - 1$
Volume input	Volume/ 10^9	$\log(1 + \text{Volume}/10^9)$
Model output	Next-day Close	Relative return $\hat{r}_{t+1}$
Final prediction	Direct output	$C_t(1 + \hat{r}_{t+1})$

The reproduced checkpoint is used for CM-v evaluation and local inference. S5-Full is the local experiment identifier for the CMamba-T architecture; it is not the published S5 layer as an independent SSM family. Four CMBlocks with $d_{state} = 16$ process its 60 day-token sequence. The latest-day representation passes through LayerNorm and Linear $32 \rightarrow 1$. The seed-23 checkpoint, selected at best epoch 321, has 57,249 parameters.

Both models are therefore scored in the same price space and on the same target dates. Because the input window, preprocessing, output parameterization, and architecture change together, this is an end-to-end model-pipeline comparison, not an architecture-only ablation.

For inference, the data are validated and the model-specific window is selected before loading the chosen checkpoint. Prediction does not update the model weights.

3.3 Metrics and Statistical Tests

For $e_t = y_t - \hat{y}_t$, the three primary metrics are

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}, \quad \text{MAE} = \frac{1}{n} \sum_{t=1}^n |e_t|, \quad (3.1)$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{e_t}{y_t} \right|. \quad (3.2)$$

Forecast direction is the sign of $\hat{y}_{t+1} - y_t$, and realized direction is the sign of $y_{t+1} - y_t$. Directional accuracy is calculated only for models that produce an upward or downward forecast; naive persistence is excluded from this metric. DM and Wilcoxon tests use the exact paired dates, are two-sided, and use significance level $\alpha = 0.05$.

To preserve serial dependence, the thesis uses a non-circular moving-block bootstrap [8, 9]. For $n = 304$, it fixes $L = \lceil n^{1/3} \rceil = 7$ as an $n^{1/3}$ -order heuristic, uses 50,000 resamples, and sets bootstrap computation seed 230,813, distinct from the training seed. The paired analysis is repeated at $L = 5$ and $L = 14$ while the data, forecasts, resample count, and computation seed remain fixed. Here L denotes bootstrap block length, not the model input window. $L = 7$ is not claimed to be optimal. The percentile 95% intervals assume approximate stationarity and short-range dependence within each split; they do not estimate retraining variation or uncertainty under future market regimes. Intervals are reported for each model's RMSE, paired RMSE differences, and directional accuracy. An exact McNemar test on paired correct/incorrect directional outcomes is supplementary [10]; the block-bootstrap interval is primary when accounting for serial dependence.

3.4 Two Trading Protocols

Paper replay executes the released Vanilla, Smart, and Extended Smart algorithms with initial capital of \$100 and zero fees. Its purpose is to audit reproduction of Table 4 in the paper; it is not an economically corrected simulation.

Corrected self-financing is a separate engine. A signal at day t uses only the current Close and the forecast for Close $t + 1$. Orders execute at the signal-day Close to preserve the same-close convention, after which positions are marked at the target Close. Fees are charged on absolute traded notional and included in position sizing. The engine maintains cash and BTC positions, tracks short collateral, limits gross exposure to 1x, and verifies

$$\text{equity}_{\text{final}} = \text{cash}_T + q_T p_T, \quad (3.3)$$

where transaction fees and borrow costs are deducted directly from cash before marking positions. The result matrix uses transaction-cost rates of 0%, 0.1%, and 0.2%; 0.2% is the thesis reporting reference. Borrow cost is set to zero in this experiment.

Chapter 4

Experimental Results

4.1 RQ1: Reproduction and Paper Reconciliation

Table 4.1 transcribes the Volume-enabled rows from Table 3 of CryptoMamba. These are aggregate values published by the authors, not results recomputed in this thesis.

Table 4.1 – Volume-enabled test results reported in Table 3 of CryptoMamba [1].

Model	RMSE	MAPE (%)	MAE	Parameters	Source
LSTM-v	2,202.1	2.896	1,668.9	204,000	Paper-reported
Bi-LSTM-v	2,080.2	2.738	1,562.5	569,000	Paper-reported
GRU-v	1,978.0	2.526	1,454.3	153,000	Paper-reported
iTransformer-v	1,905.9	2.540	1,387.47	201,000	Paper-reported
S-Mamba-v	1,651.6	2.215	1,209.7	330,000	Paper-reported
CryptoMamba-v	1,598.1	2.034	1,120.7	136,000	Paper-reported

On the paper’s complete 350-day target set, the official checkpoint evaluated through the local pipeline obtains RMSE/MAE/MAPE of 1,598.09/1,120.66/2.034%, nearly identical to the CryptoMamba-v row. The reproduced checkpoint obtains 1,612.35/1,132.54/2.049%. The corresponding gaps are 0.892%, 1.057%, and 0.718%. These differences quantify the forecasting reproduction in RQ1. The 350-day result is reported separately because it does not share the S5-Full target set.

4.2 RQ2: Controlled Evaluation on Common Dates

Tables 4.2 and 4.3 use exactly 304 dates for each model. No interpolation or metric combination across different date ranges is performed.

The following tables are 304-day subsets aligned with S5-Full and must not be compared directly with the paper’s 350-day aggregate in Section 4.1.

Table 4.2 – Controlled validation subset, $n = 304$, 16-11-2022 to 15-09-2023.

Model	RMSE	MAE	MAPE (%)	Dir. acc. (%)
Reproduced CM-v (14-day)	582.52	387.21	1.565	52.63
S5-Full (60-day)	557.34	373.55	1.503	49.01
Naive persistence	554.94	368.48	1.480	—

Table 4.3 – Controlled test subset, $n = 304$, 16-11-2023 to 14-09-2024.

Model	RMSE	MAE	MAPE (%)	Dir. acc. (%)
Reproduced CM-v (14-day)	1,703.51	1,228.55	2.125	57.57
S5-Full (60-day)	1,660.83	1,186.54	2.052	52.30
Naive persistence	1,657.55	1,185.06	2.049	—

Note: Naive persistence predicts the current closing price and therefore does not produce an upward or downward directional forecast.

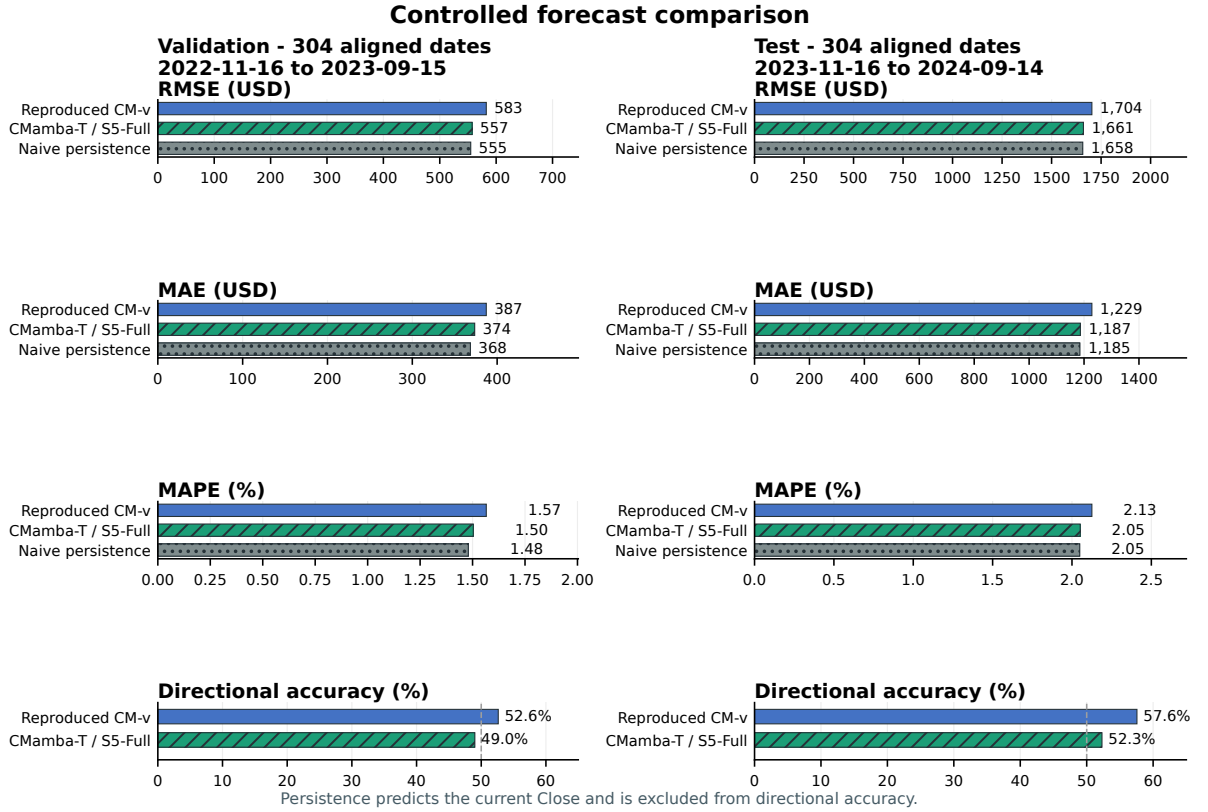


Figure 4.1 – Controlled validation and test comparison on 304 aligned target dates per split. Directional accuracy is reported only for CM-v and S5-Full.

The point estimates show lower error for S5-Full than CM-v on both splits, while persistence remains lower than S5-Full. CM-v has the highest test directional accuracy. Price-level error and directional prediction measure different properties and must be interpreted separately.

Table 4.4 – Moving-block-bootstrap 95% confidence intervals for RMSE. Δ RMSE is S5-Full minus CM-v.

Split	CM-v	S5-Full	Persistence	Δ RMSE [95% CI]
Val	[490.30, 684.89]	[477.58, 644.61]	[474.90, 642.26]	−25.18 [−55.69, 6.54]
Test	[1,480.46, 1,945.62]	[1,420.73, 1,918.17]	[1,419.78, 1,914.65]	−42.69 [−114.49, 23.44]

Table 4.5 – Sensitivity of paired 95% intervals to bootstrap block length. Negative Δ RMSE favors S5-Full; positive Δ Direction favors CM-v.

Split	L	Δ RMSE [95% CI]	Δ Direction [95% CI, pp]
Val	5	[−55.80, 5.95]	[−3.29, 10.86]
Val	7	[−55.69, 6.54]	[−2.96, 10.86]
Val	14	[−53.02, 4.23]	[−2.30, 10.20]
Test	5	[−116.46, 23.53]	[−3.62, 12.83]
Test	7	[−114.49, 23.44]	[−3.29, 12.50]
Test	14	[−106.55, 16.23]	[−2.63, 11.51]

Every interval in Table 4.5 includes zero. Thus neither paired conclusion changes for $L = 5, 7, 14$: no RMSE or directional advantage is statistically confirmed. This is a bounded sensitivity check over three dependence settings, not a claim for arbitrary block lengths.

RMSE penalizes error magnitude, whereas directional accuracy compares only the sign of change. On

test, the median absolute predicted return is 0.67% for CM-v and 0.12% for S5-Full; staying closer to the current price can reduce S5-Full’s squared error without improving direction. On validation, the 95% directional-accuracy intervals are [47.37, 59.21]% for CM-v and [44.41, 54.28]% for S5-Full; their difference is 3.62 percentage points with interval $[-2.96, 10.86]$. On test, the corresponding intervals are [51.32, 62.83]%, [47.70, 57.24]%, and a 5.26-point difference with interval $[-3.29, 12.50]$. Both paired intervals include zero. Exact McNemar tests did not reject the null of equal marginal directional accuracy (validation $p = 0.400$; test $p = 0.224$); this is not evidence of equivalence.

Table 4.6 – Descriptive chronological-half comparison; each period contains 152 consecutive dates.

Period	CM-v RMSE	S5 RMSE	Persistence RMSE	CM-v Dir. (%)	S5 Dir. (%)
Val H1	599.05	561.60	559.57	50.66	48.68
Val H2	565.49	553.04	550.27	54.61	49.34
Test H1	1,693.50	1,669.26	1,672.69	59.21	53.29
Test H2	1,713.47	1,652.35	1,642.26	55.92	51.32

Val H1/H2 cover 16-11-2022–16-04-2023 and 17-04-2023–15-09-2023; Test H1/H2 cover 16-11-2023–15-04-2024 and 16-04-2024–14-09-2024. S5-Full has lower RMSE than CM-v in all four periods, but persistence is lower than S5-Full in three; CM-v has higher directional accuracy in all four. This is a descriptive check on the same checkpoints, not four training runs or four independent holdouts.

Table 4.7 – Paired tests on 304 common dates. DM uses squared error with HAC lag 1; Wilcoxon uses absolute error.

Split	Comparison A vs B	DM p	Wilcoxon p	Lower MSE	Lower MAE
Val	CM-v vs persistence	0.106	0.024	persistence	persistence
Val	S5-Full vs persistence	0.161	0.017	persistence	persistence
Val	S5-Full vs CM-v	0.128	0.094	S5-Full	S5-Full
Test	CM-v vs persistence	0.188	0.096	persistence	persistence
Test	S5-Full vs persistence	0.455	0.977	persistence	persistence
Test	S5-Full vs CM-v	0.224	0.117	S5-Full	S5-Full

On validation, Wilcoxon identifies lower absolute error for persistence than for CM-v ($p = 0.024$) and S5-Full ($p = 0.017$); DM does not reject equal squared-error accuracy. On test, all six p -values exceed 0.05. Therefore, S5-Full’s lower RMSE than CM-v is an observed point estimate, not evidence of a statistically significant advantage.

4.3 RQ3: Paper Replay and Corrected Self-Financing

Table 4.8 preserves the paper’s zero-cost trading protocol. The official checkpoint reproduces the three published test rows almost exactly. The reproduced checkpoint generates a different action sequence, so its ending balance is lower despite only a small difference in aggregate forecast error.

Table 4.8 – Paper replay on test with \$100 initial capital and zero fees.

Checkpoint	Strategy	Paper balance	Local balance	Paper MDD (%)	Local MDD (%)
Official	Vanilla	246.58	246.58	23.48	23.48
Official	Smart	213.20	213.20	13.37	13.37
Official	Extended Smart	262.78	262.78	11.09	11.09
Reproduced	Vanilla	246.58	174.94	23.48	23.94
Reproduced	Smart	213.20	165.25	13.37	10.20
Reproduced	Extended Smart	262.78	204.26	11.09	11.74

Table 4.9 is a separate evaluation surface: 304 common dates, a self-financing engine, and a 0.2% fee. Its balances should not be compared directly with Table 4.8 because the date sets and accounting rules differ.

Table 4.9 – Corrected self-financing test results with a 0.2% fee and \$100 initial capital. Turnover and fees are in USD; every reconciliation error is zero.

Model	Strategy	Final	ROI (%)	MDD (%)	Trades	Turnover	Fees
Shared control	Buy-and-hold	158.09	58.09	26.18	1	99.80	0.20
Reproduced CM-v	Vanilla	115.97	15.97	26.29	46	5,731.07	11.46
Reproduced CM-v	Smart	95.92	−4.08	18.08	222	7,758.47	15.52
Reproduced CM-v	Extended Smart	92.11	−7.89	19.81	304	19,450.28	38.90
S5-Full	Vanilla	100.00	0.00	0.00	0	0.00	0.00
S5-Full	Smart	103.08	3.08	1.91	133	124.35	0.25
S5-Full	Extended Smart	98.39	−1.61	5.00	304	508.33	1.02
Persistence	All three modes	100.00	0.00	0.00	0	0.00	0.00

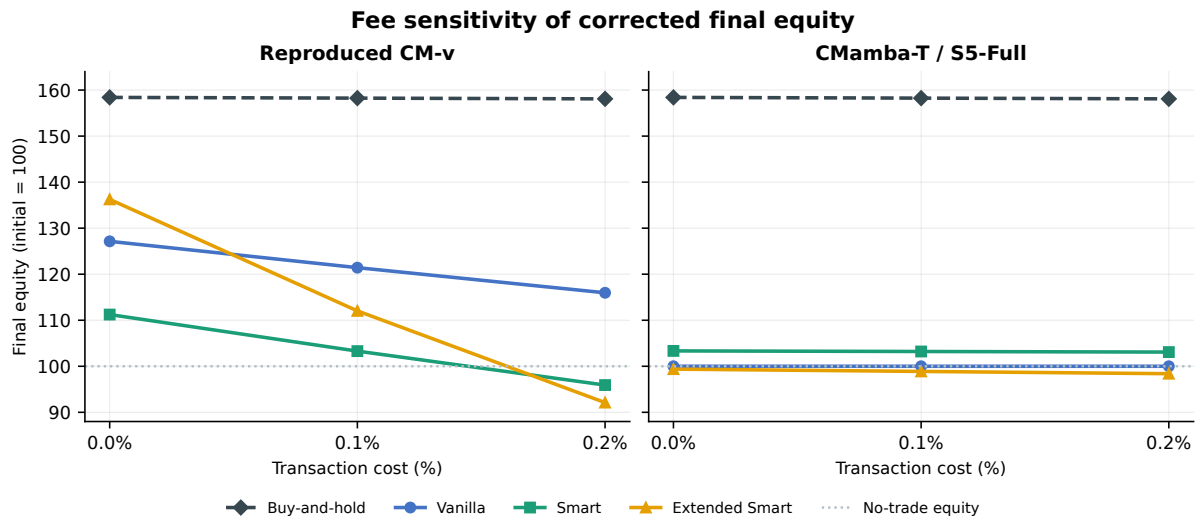


Figure 4.2 – Fee sensitivity of corrected final equity on the 304-date test split. Each panel compares Buy-and-hold, Vanilla, Smart, and Extended Smart at 0%, 0.1%, and 0.2% transaction cost.

No forecast-driven strategy outperforms buy-and-hold at any fee. As cost rises from 0% to 0.2%, CM-v Vanilla falls from 127.14 to 115.97, Smart from 111.23 to 95.92, and Extended Smart from 136.25 to 92.11. S5-Full Smart changes only from 103.34 to 103.08; Vanilla issues no trades. RQ3 therefore does not depend on choosing 0.2% alone: paper replay reproduces the released algorithm, while the corrected engine does not confirm an economic advantage, and high-turnover policies are more fee-sensitive.

Chapter 5

CMamba-T/S5-Full Experiment

5.1 Design and Scale

Both official and reproduced CM-v retain the paper’s $[B, 6, 14]$ tensor. In a separate local experiment, S5-Full receives a sequence of 60 day-tokens with a 32-dimensional embedding. Window length and model width become independent quantities, a structural difference verifiable in the source. Because the experiment changes the input window, preprocessing, output parameterization, and architecture together, it is not an architecture-only ablation.

S5-Full has 57,249 parameters, or 41.80% of CM-v, a reduction of 58.20%. Figure 5.1 places this scale beside RMSE on the same splits. Persistence is a zero-parameter reference line, not a learned-model point.

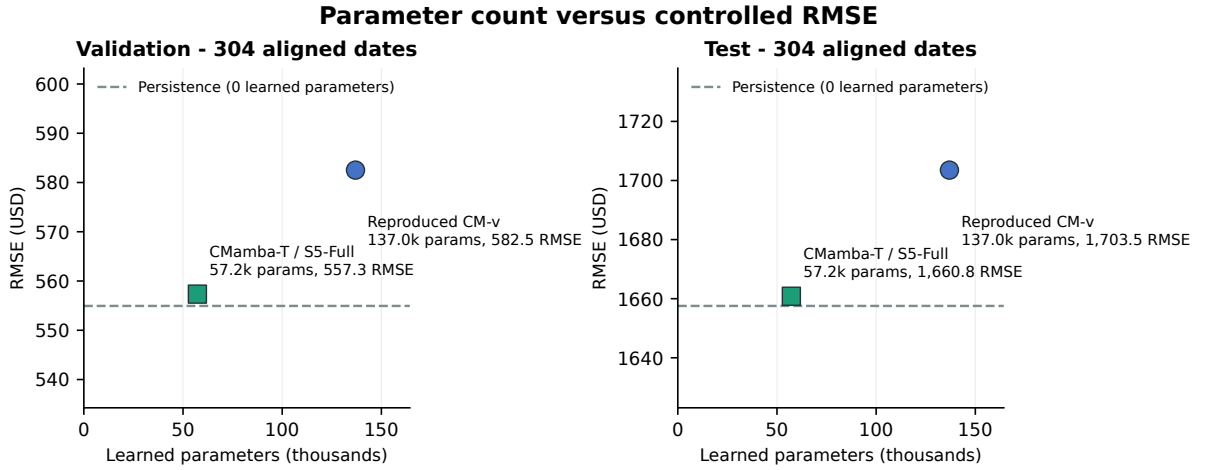


Figure 5.1 – Parameter count versus validation and test RMSE. Only CM-v and S5-Full are learned models; persistence is a zero-parameter reference.

5.2 Forecasting and Trading Results

S5-Full’s RMSE is 25.18 below CM-v on validation (557.34 versus 582.52) and 42.69 below on test (1,660.83 versus 1,703.51), while remaining 2.40 and 3.28 above persistence, respectively. Neither S5-Full–CM-v difference is significant under DM or Wilcoxon (Table 4.7).

In corrected trading, S5-Full Smart ends between \$103.34 and \$103.08 as cost rises from 0% to 0.2%; Vanilla issues no trades, and Extended Smart ends at \$98.39 at 0.2%. Lower forecast error than CM-v therefore does not imply higher profit under every policy.

5.3 RQ4 Conclusion

S5-Full has lower point-estimate RMSE than reproduced CM-v on validation and test, but not than persistence. Paired tests and intervals do not establish general superiority. Only the seed-23 checkpoint was evaluated. Forecasts, statistical tests, parameters, and trading characterize this run, not variation across training runs.

Chapter 6

Explanatory System

6.1 Architecture and Primary Workflow

The system comprises a React/TypeScript frontend, a FastAPI backend, and a Python worker from the core repository. The backend serves evaluation tables and delegates forecast requests to the worker that runs the selected checkpoint. Invalid data or an unavailable checkpoint produces an explicit error instead of a simulated result.

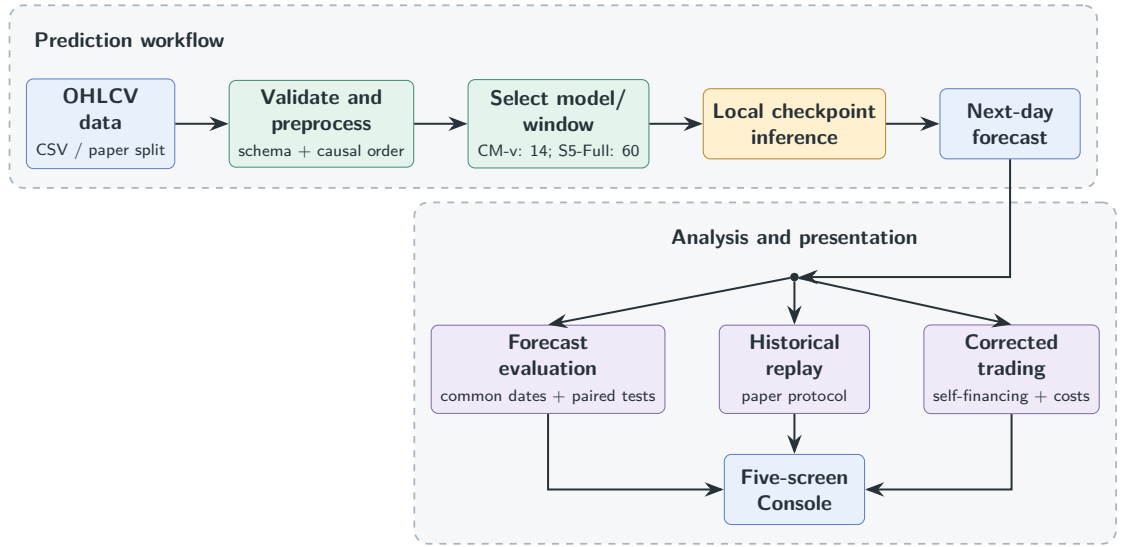


Figure 6.1 – Primary workflow from OHLCV validation and local checkpoint inference to forecast evaluation, historical replay, corrected trading, and the five-screen Console.

The primary flow is uploaded or paper data → CM-v or S5-Full selection → *local checkpoint* execution → forecast, expected return, and input window. CM-v requires at least 14 candles; S5-Full requires 60. Historical replay is an independent mode. Live HTTP appears under advanced options and sends a request only when explicitly invoked.

6.2 Five Screens

The Console retains the five-screen structure, and both the interface and embedded screenshots are in English. Table 6.1 maps each screen to its thesis role.

6.3 Functional Validation

Automated tests cover CSV parsing, window limits, both model-inference paths, invalid data, and accounting invariants. Browser checks cover all five screens, both checkpoints, valid and invalid CSV files, fewer than 60 days of history, historical replay, and both trading protocols. These checks establish prototype functionality; the thesis makes no usability claim without a user study.

Table 6.1 – Roles of the five Console screens.

Screen	Function	Displayed result
Data	Load CSV; validate schema, order, values, and history	Validated OHLCV
Evaluation	Compare reported values, local metrics, and paired tests	Forecast metrics
Predict	Select checkpoint; forecast paper or new data; separate replay	Next-day price
Trading	Switch between paper replay and corrected self-financing	Two trading protocols
Architecture	Compare the paper description, released CM-v path, and S5-Full	Model architecture

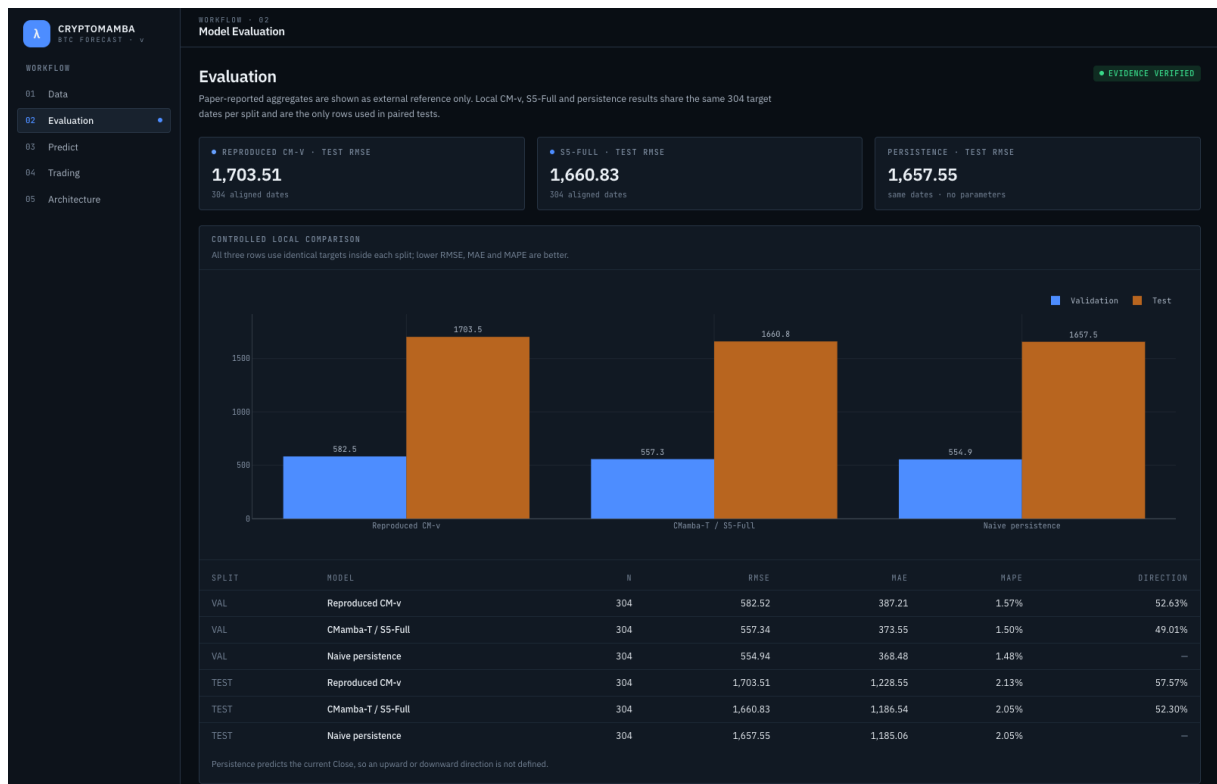


Figure 6.2 – Evaluation screen: controlled validation/test RMSE and the aligned local metric table.

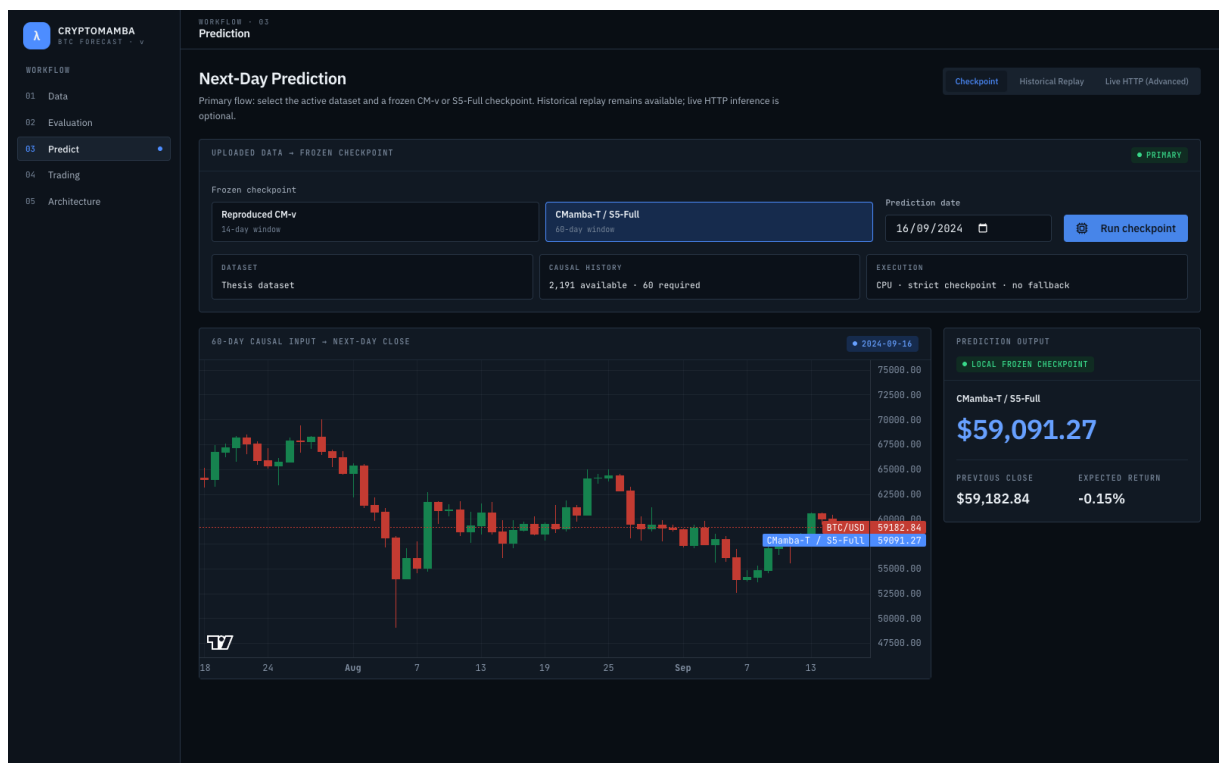


Figure 6.3 – Predict screen: uploaded or paper data is sent to the selected CM-v or S5-Full local checkpoint; the model window remains visible.

CORRECTED SELF-FINANCING • TEST • COST 0.2%

Cash, position and equity reconcile at every step. Fees are charged on traded notional; target close is used only for mark-to-market and outcome metrics.

MODEL	STRATEGY	FINAL EQUITY	ROI	MDD	SHARPE	TRADES	TURNOVER	FEES	RECONCILIATION
Reproduced CM-v	Buy & Hold	158.09	+58.09%	26.18%	1.29	1	99.80	0.200	0.0e+0
Reproduced CM-v	Vanilla	115.97	+15.97%	26.29%	0.78	46	5,731.07	11.462	0.0e+0
Reproduced CM-v	Smart	95.92	-4.08%	18.08%	-0.23	222	7,758.47	15.517	0.0e+0
Reproduced CM-v	Smart + Short	92.11	-7.89%	19.81%	-0.24	304	19,450.28	38.901	0.0e+0
Naive persistence	Buy & Hold	158.09	+58.09%	26.18%	1.29	1	99.80	0.200	0.0e+0
Naive persistence	Vanilla	100.00	+0.00%	-0.00%	0.00	0	0.00	0.000	0.0e+0
Naive persistence	Smart	100.00	+0.00%	-0.00%	0.00	0	0.00	0.000	0.0e+0
Naive persistence	Smart + Short	100.00	+0.00%	-0.00%	0.00	0	0.00	0.000	0.0e+0
CMamba-T / S5-Full	Buy & Hold	158.09	+58.09%	26.18%	1.29	1	99.80	0.200	0.0e+0
CMamba-T / S5-Full	Vanilla	100.00	+0.00%	-0.00%	0.00	0	0.00	0.000	0.0e+0
CMamba-T / S5-Full	Smart	103.08	+3.08%	1.91%	1.08	133	124.35	0.249	0.0e+0
CMamba-T / S5-Full	Smart + Short	98.39	-1.61%	5.00%	-0.40	304	508.33	1.017	0.0e+0

Figure 6.4 – Trading screen: the corrected self-financing result is shown separately from the original paper replay.

Chapter 7

Discussion and Conclusion

7.1 Answers to the Research Questions

RQ1. The official checkpoint reproduces the forecasts and paper trading replay almost exactly. On the 350-day protocol, the reproduced checkpoint differs from the reported result by 0.892% RMSE, 1.057% MAE, and 0.718% MAPE.

RQ2. On the 304 common target dates in each split, persistence has the lowest RMSE on both validation and test. S5-Full has lower RMSE than CM-v in all four chronological halves, but the paired bootstrap intervals include zero and DM does not confirm a difference. CM-v has higher directional accuracy, but neither the paired interval nor McNemar’s test confirms an advantage. These conclusions are unchanged for bootstrap block lengths 5, 7, and 14. Price-level error and direction measure different properties.

RQ3. Paper replay confirms reproduction of the released algorithms. With the self-financing engine, no forecast-driven strategy outperforms buy-and-hold at 0%, 0.1%, or 0.2% cost on the same 304 test dates. Turnover explains the fee sensitivity of frequently trading policies.

RQ4. S5-Full reduces parameter count by 58.20% and has lower RMSE than CM-v on both measured splits, but does not outperform persistence. Its model contribution is therefore a lightweight experiment evaluated across forecasting, statistical testing, and trading, not a claim of the best model.

7.2 Limitations

- The data cover one asset, one frequency, and one horizon. Rankings do not automatically generalize to other assets or periods.
- The 60-day S5-Full window was not compared with alternatives. The historical test set has already been used for reporting, so subsequent decisions require a new holdout.
- The corrected engine retains signal-day Close execution for reconciliation. This same-close abstraction must be replaced by next-open execution with observed slippage, spread, and borrow costs for a realistic evaluation.
- The Console is a local prototype. It has not undergone load testing, deployment security audit, user research, or integration with an order-management system.

7.3 Future Work

The first priority is to freeze the current protocol and evaluate it once on a holdout after 09-2024, followed by multiple S5-Full seeds and window comparison under a fixed validation protocol. Later extensions may add exogenous data, multiple assets, and additional horizons. The trading engine should adopt next-open execution with a complete cost model. Live HTTP may be added as an optional adapter after operational validation; it does not change the local-checkpoint contract.

7.4 Conclusion

This thesis does more than repeat a published result table. It separates Paper-reported values from local results, aligns every comparison on common dates, adds persistence and paired tests, reconciles paper replay, and implements a corrected self-financing backtest. S5-Full extends the work from reproduction to a lightweight model experiment, while its conclusions remain bounded by persistence and statistical significance. The five-screen Console presents the complete data–prediction–evaluation–trading–architecture chain directly from local checkpoints.

Within Vietnam’s pilot crypto-asset market, the prototype’s application value lies in result visualization and explicit experimental limitations, not in issuing investment signals.

Data and Code Availability

The original paper and source code are available at <https://arxiv.org/abs/2501.01010> and <https://github.com/MShahabSepehri/CryptoMamba>. The forecast, statistical, and trading data used in the result tables are supplied with the project materials.

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